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Database Management 112

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Lab 1: PostgreSQL setup

An example of an imagined database may include something like an online video game with different character interactions and the need for information to be stored based on each user's specific and unique information. There are a variety of elements that are stored in an online game's database, some of which include the user's account data, the connections this account has including online connections, as well as system settings and performance data. The database organizes this data into information by categorizing it and giving the data of a category certain importance and meaning. Without context and organization, data would not be able to be interpreted, for example, statistics and account information would just be numbers and letters without context. Information provides value in context because you can distinguish pieces of info from one another creating differences and giving meaning to the data.

The hierarchical data model is based off of a tree model where data is reached by going through the root of the database. The relational database systems model showed data in tables or relations with each other and was less focused on the storage structure. The shortcomings of the hierarchical model is that even very simple queries were a very consuming task to write programs for. With this in consideration, I think XML is a good way to store data because a file labeled with XML regards a large file that holds a collection of many smaller files that serves as a database and can be queried and changed. However if it is being queried a lot, it is not an optimal choice.

